

SOUMILI SARKAR

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PROFESSIONAL SUMMARY

Results-driven Computer Science undergraduate with hands-on experience in Machine Learning, Deep Learning, NLP, and Generative AI. Skilled in Python and TensorFlow, with practical exposure to transformer-based architectures (BART), LLM concepts, prompt engineering, and model fine-tuning through academic projects and an industrial Generative AI internship. Comfortable with Git/GitHub version-controlled workflows and Agile development. ServiceNow Certified Application Developer and certified in Generative AI and Data Science, with a strong CGPA of 9.06/10. Seeking an AI/ML Engineer role to build intelligent, LLM-powered systems.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C, C++, SQL

ML / AI & Generative AI: TensorFlow, Keras, Scikit-learn, Pandas, NumPy, NLP (NLTK / spaCy), Transformer Models (BART), LLM Concepts, Prompt Engineering, Model Fine-Tuning

Data Visualization: Matplotlib, Seaborn, Plotly, Streamlit, Power BI, Tableau

Web & Frontend: React.js, Next.js, HTML5, CSS3, REST APIs

Databases & Tools: MySQL, Git/GitHub, VS Code, Jupyter Notebook, Google Colab

Core Competencies: Data Preprocessing, Feature Engineering, Model Evaluation, Prompt Engineering, BI Reporting, Agile, Problem-Solving

PROJECTS

Mental Health Detection & Trend Analysis System | *Group Project* 2025

- Collaborated in a team to build an NLP-powered classification system for mental health text data, combining classical ML models (Logistic Regression, SVM, Random Forest) with a fine-tuned BART transformer model, using tokenization, stopwords removal, and TF-IDF feature extraction via NLTK.
- Engineered a weighted ensemble (transformer + classical ML) with bias-correction and keyword-flagging logic and a three-tier confidence output, improving classification robustness on Reddit-sourced text data.
- Led trend analysis module — performed time-series analysis on 5,000+ records, surfacing behavioral patterns with interactive Seaborn and Plotly dashboards, enabling data-driven insight generation.
- Coordinated model evaluation and reporting across team members, applying SHAP-style analytical insights and increasing overall classification confidence.

Diabetes Prediction System using ANN | *Personal Project* 2025

- Designed and trained a 3-layer Artificial Neural Network (ANN) using TensorFlow/Keras to predict diabetes risk from 8 clinical features, achieving ~82% classification accuracy on the Pima Indians dataset.
- Engineered a preprocessing pipeline with Pandas and NumPy, handling missing values and normalizing features — reducing model error by ~15% compared to raw input baseline.
- Implemented probability-based output with Matplotlib visualizations and deployed an interactive prediction interface using Streamlit, improving prediction interpretability for non-technical stakeholders.

Calorie Tracker Web App with AI Image Analysis | *Personal Project* 2025

- Developed a responsive full-stack web application using HTML5, CSS3, and JavaScript, integrating the Grok LLM API for real-time AI-powered food image recognition and calorie estimation — hands-on experience consuming and integrating third-party AI/LLM platforms.
- Implemented drag-and-drop and live camera capture UI features, reducing food logging time by ~60% compared to manual entry; integrated ImgBB API for seamless image hosting.
- Designed a modern, mobile-first interface with zero external CSS frameworks, achieving sub-2s load time and 100% client-side processing; managed codebase using Git/GitHub version control.

INTERNSHIPS & TRAINING

Industrial Training — Generative AI | *Ardent Computech Pvt. Ltd.* Feb 2026 – Mar 2026

- Completed intensive training on Generative AI and LLM fundamentals, including prompt engineering and fine-tuning techniques for large language models; applied learnings to design and build a prototype chatbot using Python.

- Completed 300+ hours of hands-on AI/ML training covering supervised learning, neural networks, and model deployment; implemented 5+ end-to-end ML projects in Python.

EDUCATION

B.Tech in Computer Science & Engineering

Aug 2023 – Present

Narula Institute of Technology, Kolkata

CGPA: 9.06 / 10.0 (up to 5th Semester)

ISC Class XII (Science)

2021 – 2023

Pearls of God School, Hooghly

Percentage: 82%

CERTIFICATIONS & ACHIEVEMENTS

- ServiceNow Certified Application Developer (CAD) | June 2026
- Cloud Computing - NPTEL (Score: 85%) | May 2026
- Introduction to Data Science — Certified | Mar 2026
- Programming in Java — NPTEL (Score: 94%) | Nov 2025
- Certificate of Merit — Naukri Campus Young Turks 2025 (96.27 percentile) | Oct 2025
- Young Professional Course — TCS iON (AI, Business Etiquette, Communication) | Jul 2025